

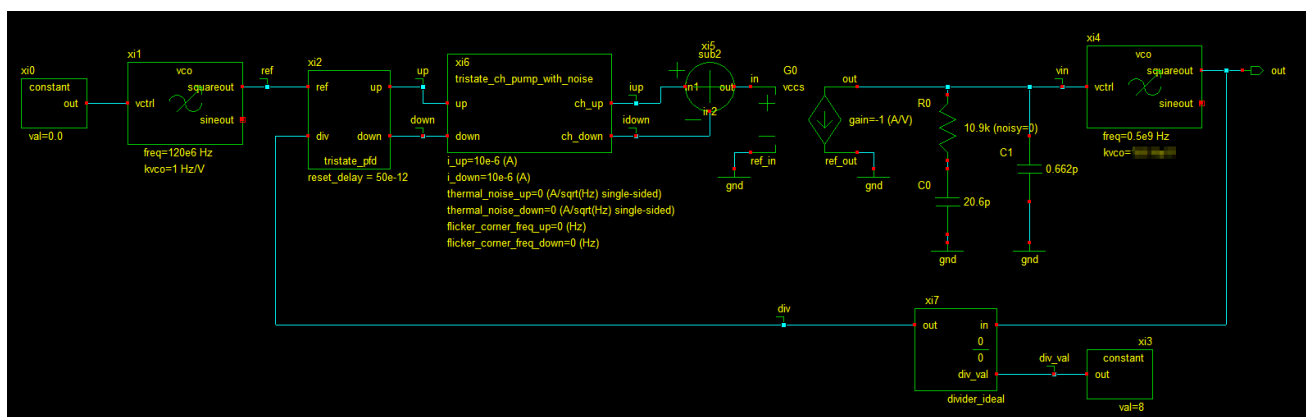
Analog/Mixed-Signal Simulation and Modeling**Lab 06****PLL System-Level Design and Simulation****Objectives**

1. Learn how to use fast system level simulation tools.
2. Learn how to design and simulate a PLL system.

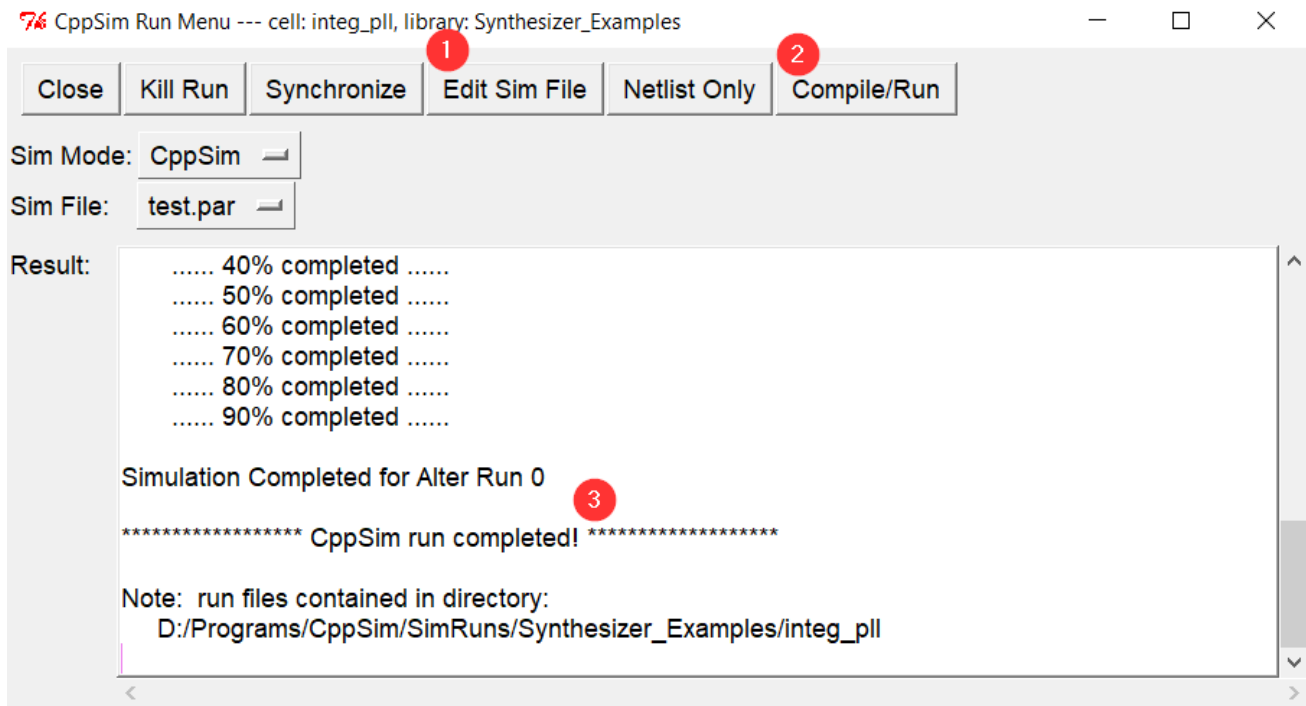
Instructions

1. Download and install CppSim: <https://www.cppsim.com/download.html>
Note: You must install CppSim in a path that does NOT have spaces.
2. From the Windows start menu: Run Sue2 (CppSim schematic editor).
3. Draw the schematic of an integer-N PLL similar to the one in “pll_design.pdf” and adjust all settings as shown in the figure below.

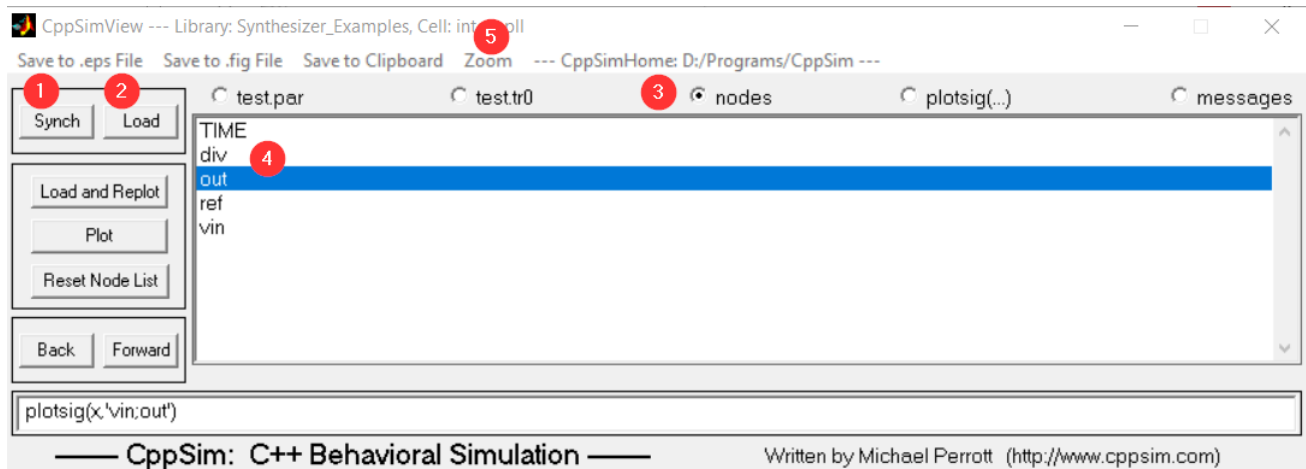
Hint: Instead of creating a design from scratch, you may edit one of the existing Synthesizer Examples and use (File -> Save As) to save it to a new file.



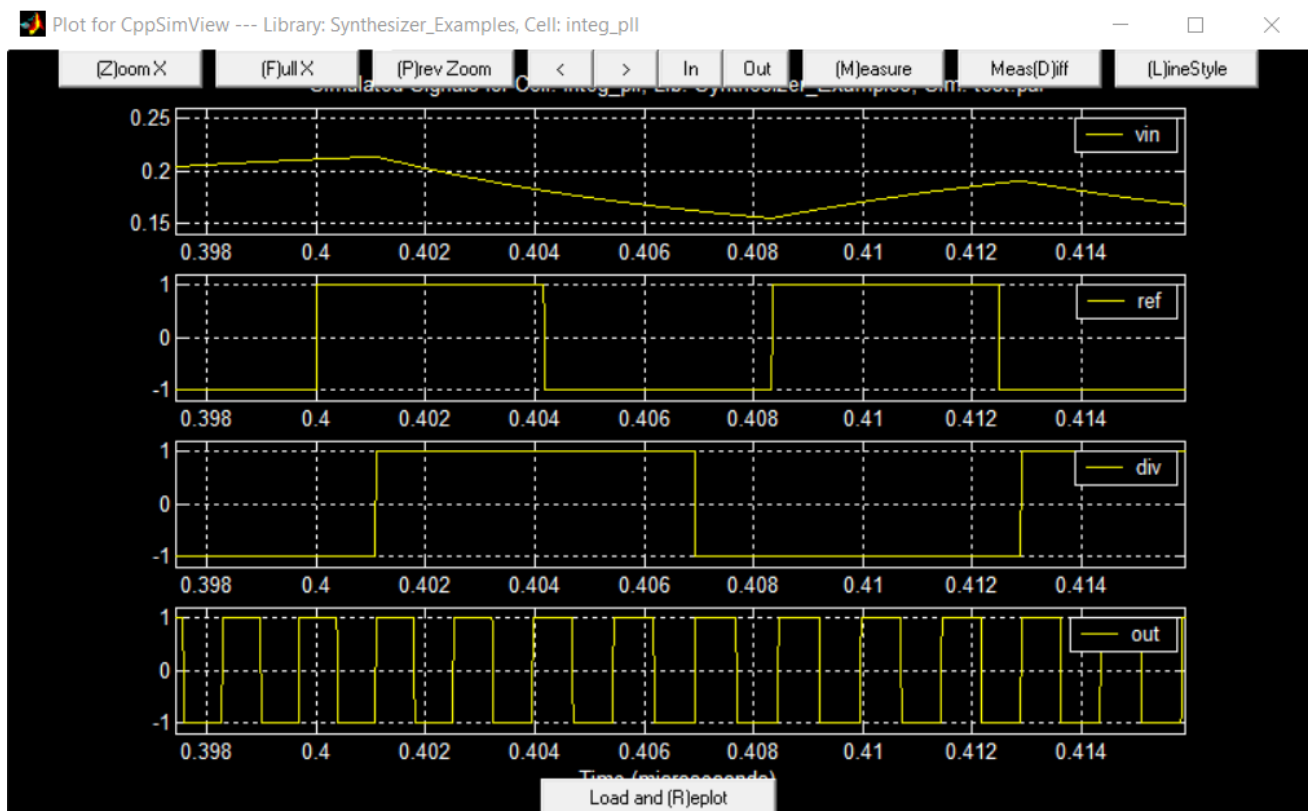
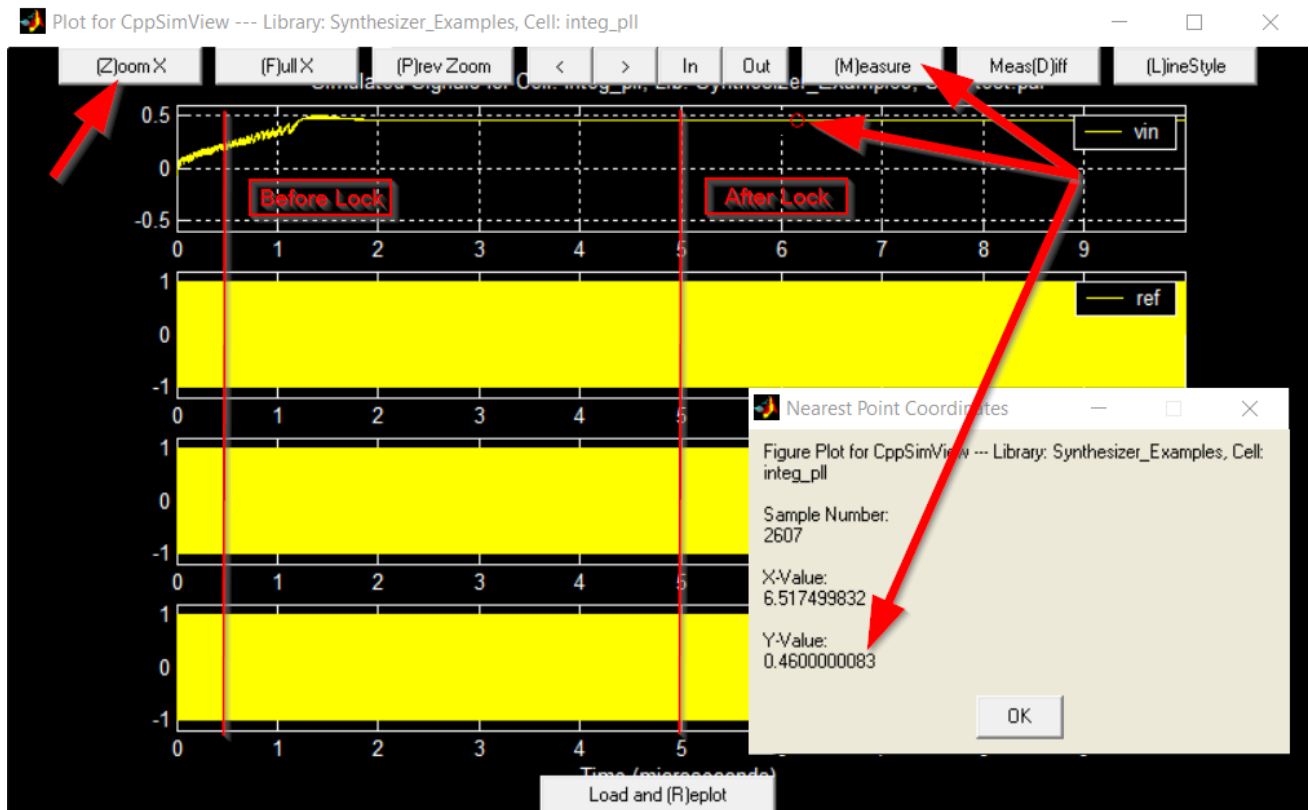
4. The parameter kvco is hidden in the above figure because you will use a unique random number. To calculate your kvco, visit this random number generator website to get a random number between 50 and 400: <https://www.calculator.net/random-number-generator.html?slower=50&supper=400&ctype=1&s=1922&submit1=Generate>
5. Use $kvco = \text{your_random_num} / 100 * 1e9 \text{ Hz/V}$
6. Open CppSim Run Menu (Tools -> CppSim Simulation).
7. Choose “Edit Sim File” and set the following settings:
num_sim_steps: 1e6
Ts: 10e-12
probe: ref div vin out
global_nodes: gnd=0.0 avdd=1.2 dvdd=1.2

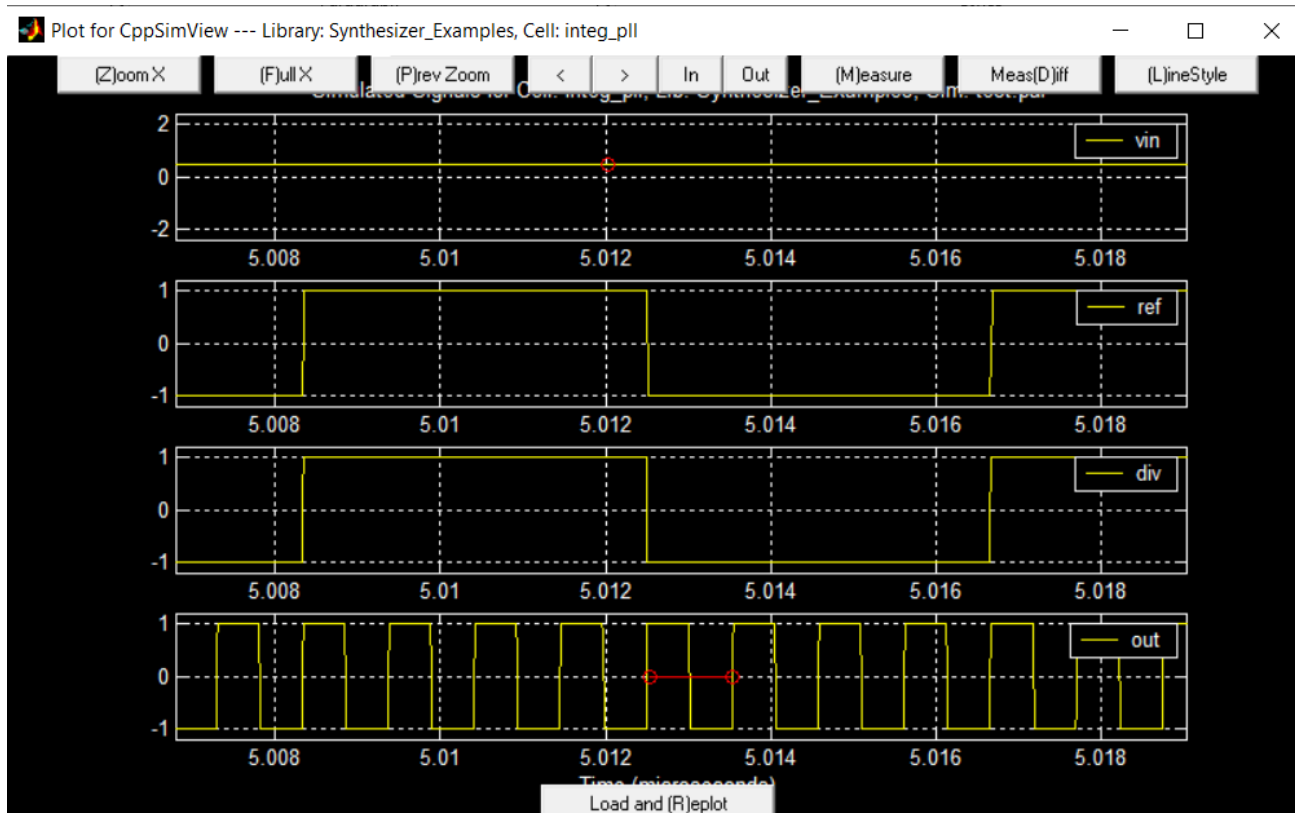


8. Choose “Compile/Run”. Make sure the simulation is completed without errors as shown above. Debug the errors if any.
9. From the Windows start menu: Run CppSimView (the waveform viewer).
10. Click “Synch”, then “Load”, then “nodes”, then double click each of the four nodes in the following order (vin, ref, div, vout). Then click “Zoom”.



11. Add a cursor using “Measure” to show the value of vin at lock clearly as shown below. Use “Zoom X” to zoom in the area before lock and after lock. Report snapshots to clearly show that ref and div are out of phase then in phase as shown below.





Part 1

Index	Deliverable
1.	Schematic of the PLL drawn in Sue2.
2.	Snapshot of the random number generation website showing your random number. Calculate k_{vco} .
3.	Given k_{vco} , analytically calculate vin at which the PLL will achieve lock.
4.	Snapshot of CppSim Run dialog showing successful simulation completion.
5.	Snapshot of the simulation results with a probe showing vin value as shown above.
6.	Compare the simulated vin value with the analytically calculated value. Comment.
7.	Clear Zoom X snapshot before lock as shown above.
8.	Clear Zoom X snapshot after lock as shown above.

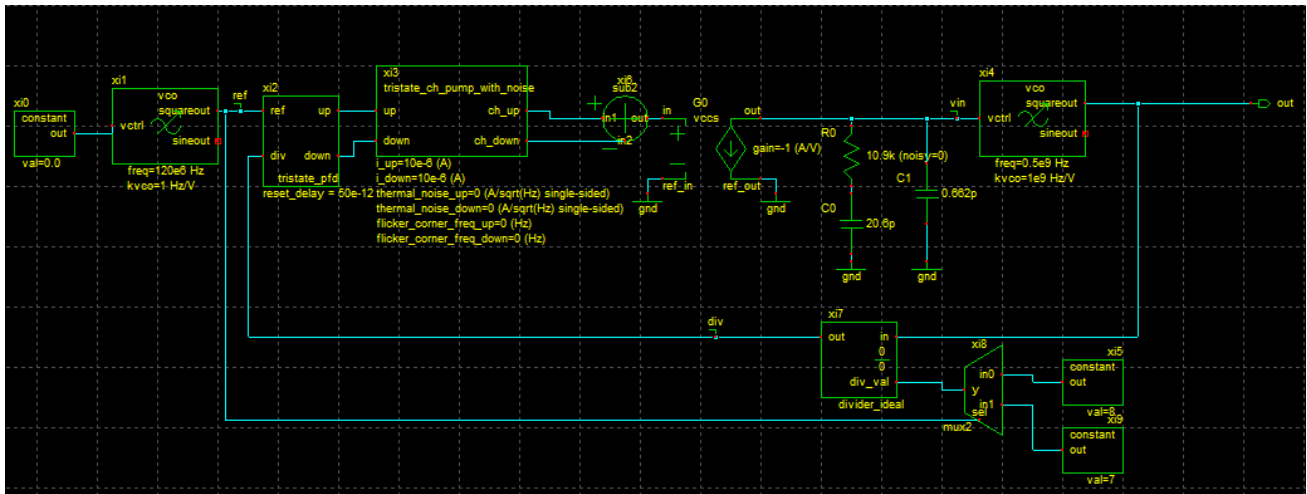
Part 2

Add a name to the charge pump output (output of xi5 sub2 in the above schematic). Use "Edit Sim File" to probe (up,down,chp_out) signals.

Index	Deliverable
1.	Plot (ref,div,up,down). Report clear Zoom X snapshot showing the PFD operation.
2.	Reset node list. Plot (vin,chp_out).
3.	[Optional] In Sue2 click (Doc -> PLL Design Assistant Manual). From the Windows start menu, run PLLDesign tool. Enjoy 😊

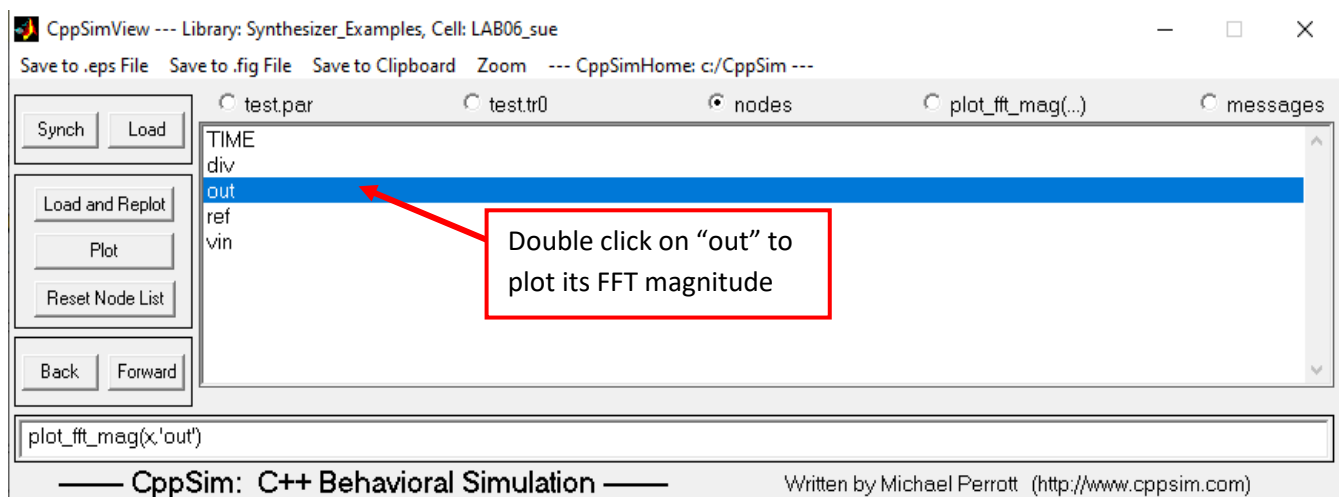
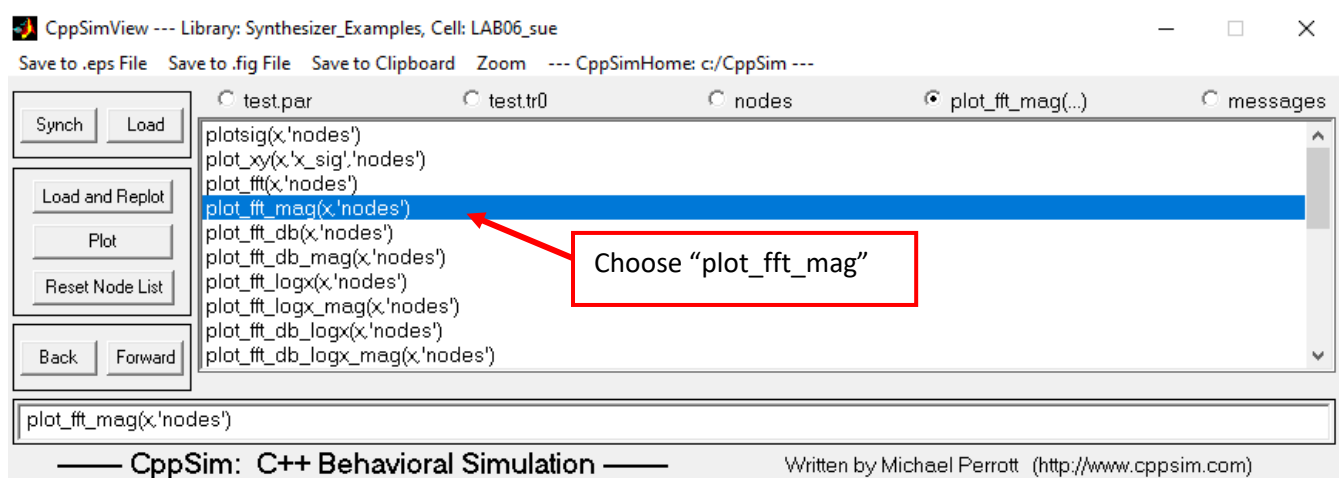
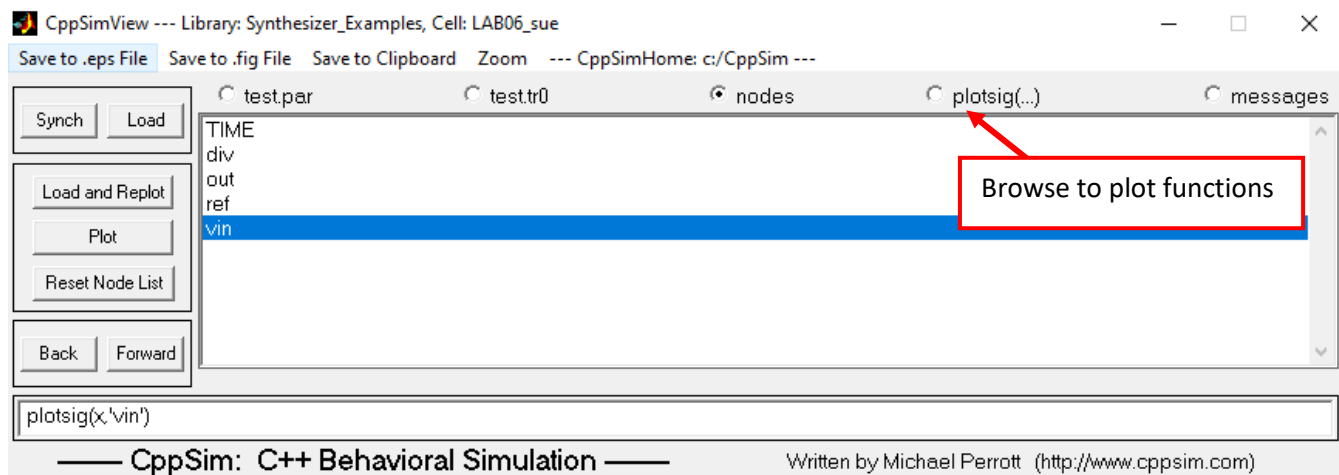
Part 3

Modify your PLL to obtain a 900MHz clock after locking. Using multiplexer-2 from CppSimModules (use reference clock as the selector) to divide over two different values using the same divider.



Index	Deliverable
1.	Given k_{vco} and the new divider value, analytically calculate v_{in} at which the PLL will achieve lock.
2.	Snapshot of the simulation results showing v_{in} waveform after locking. Comment.
3.	Use CppSimView plot functions to plot the output spectrum (FFT). Comment. For help, in Sue2 click (Doc -> CppSim/VppSim Primer).

Hint: How to plot FFT



Thanks to all who contributed to these labs. If you find any errors or have suggestions concerning these labs, please contact Hesham.omran@eng.asu.edu.eg.